

xiii. The Comets and Meteors

*A comet is an icy, small solar system body that, when passing close to the sun, warms and begins to release gases. This produces a visible atmosphere or coma and sometimes also a tail. Comets usually have highly elliptical orbits and they revolve around the sun. Sometimes the tail of these comets can be millions of kilometers long. *Comets generally have two kinds of tails – (1) comprised of dust particles and (2) ionized. Both are generated from the Coma. *The sun's radiation pressure and solar wind accelerate materials away from the comet's head at different velocities according to the size and mass of the materials. ***Halley** comet visits the planet Earth at an interval of every approx. 76 years. The Halley is a short period comet. On 23rd July 1995, the Hale-Bopp comet was seen near Jupiter. It was discovered by Alan Hale (New Mexico) and **Thomas Bopp** (Arizona). The last names of these two people were combined to name this comet. The Hale-Bopp is a long period comet which takes 2534 years to revolve around the Sun. *The Hale-Bopp comet is 1000 times brighter than the Halley comet, even though they are at the same distances. The comet **Shoemaker Levy-9** was discovered on 24th March, 1993 by astronomers Eugene M. Shoemaker, Carolyn Shoemaker and David Levy. The last names of both the astronomers were used to name the comet. Between 16 and 22 July 1994, the comet collided with Jupiter, providing the first direct observation of an extraterrestrial collision of solar system objects. However, Jupiter was not harmed by this collision.

*67 P/Churyumov-Gerasimenko was the destination of the **Rosetta mission** launched on 2nd March 2004. It was the first mission to land a **spacecraft on a comet**. It was launched from the Guiana Space Centre in French Guiana on an Ariane 5G+ rocket and reached comet Churyumov-Gerasimenko on 6 August 2014.

1. The tail of comet is directed away from the Sun, because–

- (a) As the Comet rotates around the Sun, the lighter mass of Comet is pushed away due to centrifugal force alone.
- (b) As the Comet rotates, the lighter mass of the Comet is attracted by some stars situated in the direction of

its tail.

- (c) The radiation emitted by the Sun exerts a radial pressure on the comet throwing its tail away from the Sun.
- (d) The Tail of the Comet always exists in the same orientation.

I.A.S. (Pre) 1997

Ans. (c)

In the outer solar system, comets remain frozen and are extremely difficult or impossible to detect from Earth due to their small size. As a comet approaches the inner solar system, solar radiation causes the volatile material within the comet to vaporize the steam out of the nucleus, carrying the dust away with it. The streams of dust and gas thus released form a huge, extremely tenuous atmosphere around the comet called the coma and the force exerted on the coma by the Sun's radiation pressure causes an enormous tail to form, which points away from the Sun. Hence (c) is the correct answer.

2. Hale-Bopp is the name of a–

- (a) Comet
- (b) Cartoon Character
- (c) International Company
- (d) Toy

M.P.P.C.S. (Pre) 1997

Ans. (a)

Hale-Bopp is a long period comet that was discovered in 1995. It was shining near the orbit of Jupiter around 23rd July, 1995. This comet was discovered by Alan Hale (New Mexico) and Thomas Bopp (Arizona) so it was named by joining the last names of the two astronomers.

3. With which planet comet Shoemaker-Levy collide –

- (a) Pluto
- (b) Mars
- (c) Jupiter
- (d) Saturn

I.A.S. (Pre) 1995

Ans. (c)

Comet Shoemaker-Levy-9 was discovered on 24th March 1993 when astronomers Carolyn Shoemaker, Eugene M. Shoemaker and David Levy captured its glimpse in their Schmidt telescope. During 16-22 July, 1994 it collided with the massive planet Jupiter. In this collision no harm was done to Jupiter. It was the first event of collision, of two components of the solar system seen from the Earth.

4. Meteor is a –

- (a) Fast moving star
- (b) A piece of matter which has entered the earth atmosphere from outer space